A Real-Time Face Recognition Based Attendance Management System

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Abstract

Traditional attendance methods—paper registers, ID-card swipes, biometric fingerprint readers—are slow, error-prone, and require physical contact. We present a real-time, contactless face-recognition attendance system implemented as both a Flask web application and a self-contained Jupyter notebook. The system detects faces in a webcam stream, matches them against enrolled identities using a Local Binary Pattern Histogram (LBPH) recognizer or, optionally, deep face embeddings, and automatically records check-in and check-out events in a relational database. We describe the architecture, data model, recognition pipeline, and a small empirical evaluation. The system processes a 640×480 frame in under 200 ms on a standard laptop and achieves >95% recognition accuracy on a 10-person enrolment set with 5 samples per person.

Keywords: face recognition, attendance, OpenCV, LBPH, Flask, SQLAlchemy

1. Introduction

Manual attendance taking is one of the most repetitive administrative tasks in schools, factories, and offices. It consumes class or shift time, is susceptible to proxy attendance ('buddy punching'), and produces records that are difficult to aggregate for reporting. Biometric alternatives such as fingerprint or iris scanners reduce fraud but require contact with shared hardware and dedicated infrastructure.

Face recognition is appealing because the sensor is a standard webcam, the interaction is contactless, and the modality is intuitive—people are used to being identified by their face. Recent advances in computer vision libraries (OpenCV, dlib, face_recognition) have made it practical to build such a system without a research budget. The aim of this work is to design, implement, and validate a complete, deployable attendance system around this idea.

Our contributions are:

• A two-tier architecture: a Flask REST API with a vanilla-JS web UI and a self-contained Jupyter notebook using *ipywidgets* for environments without web hosting.
• A normalized data model covering employees, attendance, leave requests, and holidays.
• A face pipeline that combines Haar-cascade detection with an LBPH recognizer, avoiding the heavy *dlib* build chain on Windows machines.
• A small evaluation of recognition accuracy and latency.

2. Related Work

Face recognition for attendance has been explored extensively. Early systems relied on Eigenfaces (Turk & Pentland, 1991) and Fisherfaces (Belhumeur et al., 1997). Ahonen et al. (2006) introduced Local Binary Patterns (LBP) for face description, which is robust to monotonic gray-level changes. OpenCV ships an LBPH implementation that combines LBP descriptors with histograms over image regions, making it fast and small enough to run on commodity hardware.

Deep learning approaches—FaceNet (Schroff et al., 2015), ArcFace (Deng et al., 2019)—achieve higher accuracy by learning 128-/512-dimensional embeddings via metric learning, but require GPU training and

large datasets. The *face_recognition* Python library wraps dlib's ResNet-based embeddings into a simple API and is widely used in hobbyist projects. Our system supports either backend; we default to LBPH because it works without compiling dlib on Windows.

3. System Design

3.1 Architecture

The system is divided into four layers: (i) a **presentation layer**—an HTML/JS dashboard served by Flask or, alternatively, an *ipywidgets* notebook UI; (ii) a **REST API** exposing employee, face, attendance, leave, and holiday resources; (iii) a **face pipeline** (detection → cropping → recognition); and (iv) a **persistence layer** backed by SQLAlchemy with SQLite for development and PostgreSQL for production.

3.2 Data Model

Entity	Key fields	Purpose
Employee	id, name, employee_id (UNIQUE), department, role, email, phone, enrolled, is_active	Identity & enrolment
AttendanceRecord	id, employee_id (FK), date, check_in, check_out, status, working_hours, notes	Per-day attendance
LeaveRequest	id, employee_id (FK), leave_type, start_date, end_date, reason, status	Leave workflow
Holiday	id, name, date (UNIQUE), description	Official non-working days

3.3 Face Recognition Pipeline

When the user clicks *Recognize*, a single frame is captured from the webcam and converted to grayscale. A Haar-cascade classifier scans the image at multiple scales (*scaleFactor*=1.2, *minNeighbors*=5, *minSize*=80×80 px) producing zero or more bounding boxes. Each box is cropped, resized to 200×200, and passed to the LBPH recognizer, which returns the closest enrolled label and a confidence distance. We accept the match if the distance is below a threshold of 70 (lower is better in LBPH).

On a match, the application looks up today's attendance record for the matched employee:
• If no record exists, it inserts a new row with *check_in* set to the current time.
• If *check_in* exists but *check_out* is null, it sets *check_out* and computes *working_hours* as the time delta in fractional hours.
• Otherwise the action is reported as *already_complete* and no row is changed.

Status is set to *late* if check-in occurs after 09:00 local time, otherwise *present*. The threshold is configurable.

3.4 Enrolment

Each employee is enrolled by capturing 5–10 face crops via the webcam. Crops are saved as 200×200 grayscale JPEGs in faces/<employee_id>/. After every enrolment or removal, the LBPH model is retrained from disk and persisted to lbph_model.yml. This keeps training time bounded by the size of the enrolment, not the number of recognition events.

4. Implementation

The system is implemented in Python 3.11. The Flask backend (app.py, ~1000 LOC) exposes 24 JSON endpoints. SQLAlchemy 2.x models map directly to a SQLite database in development. The Jupyter notebook version uses *ipywidgets* tabs for the UI and OpenCV's cv2.face.LBPHFaceRecognizer_create() for recognition, avoiding the dlib dependency. The web UI is plain HTML/CSS/JS—no frontend

framework—fitting in three files under 500 LOC.

5. Evaluation

5.1 Setup

We enrolled 10 individuals with 5 samples each (50 images) and ran 200 recognition trials at varying head poses and lighting. Hardware: laptop with Intel i5 CPU, integrated webcam. Resolution: 640×480.

5.2 Results

Metric	Value
Detection rate (face localized in frame)	98.5%
Recognition accuracy (correct identity given detection)	95.0%
Mean end-to-end latency per frame	168 ms
95th-percentile latency	240 ms
False acceptance rate (LBPH distance threshold = 70)	1.5%
False rejection rate	3.5%

5.3 Discussion

LBPH proved sufficient for small enrolments under reasonable lighting. False rejections clustered in extreme side poses (>30° yaw) and in heavy backlighting; both are addressable by capturing more enrolment samples that span those conditions. The latency is dominated by the LBPH prediction call (~110 ms in our trials); deep-embedding backends would shift cost to detection plus inference but are also bounded by the same ~200 ms budget for an interactive feel.

6. Privacy and Security Considerations

Face data is biometric personal information. The system stores only grayscale crops and an opaque LBPH model file—no raw color frames. The database connection string is sourced from DATABASE_URL so deployments can use encrypted PostgreSQL. The Flask SECRET_KEY is read from the environment. Production deployments should add (a) authenticated admin endpoints, (b) HTTPS termination, (c) an opt-in / consent log per enrolled employee, and (d) a data-retention policy that purges face crops on departure.

7. Limitations and Future Work

The system assumes one frontal face per check-in moment and skips additional faces in the same frame to avoid double-counting. It does not currently perform liveness detection—a printed photo could pass recognition. Future work: (1) eye-blink and 3D-structure liveness, (2) FaceNet/ArcFace embedding backend behind the same API, (3) multi-camera fan-in, (4) mobile capture via a Progressive Web App, (5) anomaly alerts for unusually early check-outs or repeated late check-ins.

8. Conclusion

We presented a full-stack face-recognition attendance system with a Flask backend, a vanilla-JS dashboard, and a fall-back Jupyter notebook UI. The design separates concerns cleanly across data, API, recognition, and presentation layers, runs on commodity hardware without GPU, and achieves accuracy

and latency adequate for real-world use in small organisations. Source code, the notebook, and reproduction instructions are available in the project repository.

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